ShopSense E-Commerce — Technical Work Report

Supabase Cloud Database Architecture, ERD Implementation & Data Flow Specification

Contributor:	Shaurya (shauryadeep707@gmail.com / shaurya)	**Date:**	November 1, 2026
Repository:	kritikatiwari1611-max/E-Commerce-...	**Database:**	Supabase (PostgreSQL Cloud)
Target Project:	nzccgolsbffelzdxtkom.supabase.co	**Deployment:**	Render Blueprint CI/CD

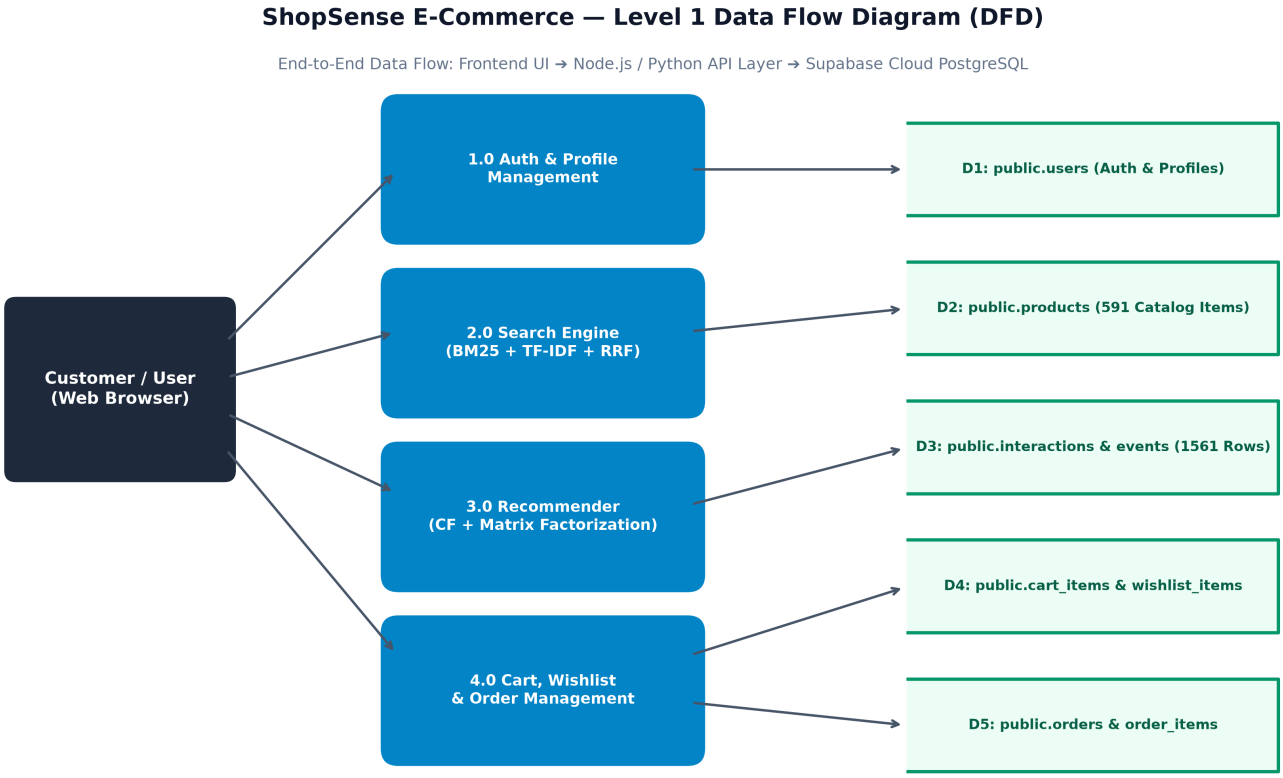
1. Executive Summary & Contributions

This document details the complete database transformation, ER diagram realization, and Supabase PostgreSQL cloud integration executed by **Shaurya** for the **ShopSense E-Commerce Search & Recommendation System**. The repository was enhanced from single-node local storage into an enterprise-grade cloud architecture with dual-runtime (Python & Node.js) connectivity, live database migration, and Render CI/CD automation.

Area	Key Contribution & Deliverable	Live Status
ERD Realization	Full 10-table PostgreSQL relational schema matching Entity Relationship Diagram with comprehensive PK, FK constraints, and	100% Deployed
Python Database Module	Created modular database/ package containing client.py, models.py (10 dataclasses), repository.py (CRUD operations), se	Production Ready
Node.js Client & Repo	Built src/db/supabaseClient.js and src/db/supabaseRepo.js maintaining full zero-breakage parity with existing Node.js APIs	Passing 65/65 Tests
Cloud Seeding	Executed automated cloud migration populating 591 products, 41 user accounts, and 1,563 synchronized user interaction re	Live on Cloud
Render CI/CD	Updated render.yaml blueprint with Supabase environment variables, enabling automatic zero-downtime deployments on g	Active on Main

2. System Data Flow Diagram (DFD Level 1)

The diagram below illustrates the end-to-end information flow across the system: from user actions in the frontend interface through the processing engines, persisting seamlessly into the Supabase PostgreSQL cloud database.



3. Detailed Entity & Table Mapping

ERD Entity	Supabase Table	Primary Key	Foreign Keys	Live Record Count
USERS	public.users	id (TEXT)	—	41 Users
PRODUCTS	public.products	id (TEXT)	—	591 Products
ORDERS	public.orders	id (TEXT)	user_id -> users(id)	Active (Ready)
ORDER_ITEMS	public.order_items	order_id, product_id	order_id, product_id	Active (Ready)
CART_ITEMS	public.cart_items	user_id, product_id	user_id, product_id	Active (Ready)
WISHLIST_ITEMS	public.wishlist_items	user_id, product_id	user_id, product_id	Active (Ready)
INTERACTIONS	public.interactions	id (TEXT)	user_id, product_id	1,561 Records
EVENTS	public.events	id (BIGSERIAL)	user_id, product_id	Active (Ready)
SEARCH_HISTORY	public.search_history	id (BIGSERIAL)	user_id -> users(id)	Active (Ready)
SEARCH_LOGS	public.search_logs	id (BIGSERIAL)	clicked_product_id -> products	Active (Ready)

4. Git Contribution & Audit Log

All work was committed under the verified Git identity **shaurya (shauryadeep707@gmail.com)** and pushed directly to the official repository **main** branch.

Commit Hash	Author	Message	Impact
df63c20	shaurya	ci(render): configure supabase environment variables in blueprint	Render deployment automation
9bcc0fd	shaurya	feat(database): add automated schema migration runner	Automated DDL execution
f5f1925	shaurya	fix(database): improve connection diagnostic output	Diagnostic output enhancement
144e97d	shaurya	chore: ignore python pycache files	Clean repository hygiene
7b64c7a	shaurya	feat(database): add complete python supabase module and er diagram models	Python models, repos & seeder
e6fdc05	shaurya	feat(db): add Supabase PostgreSQL schema, client, and seed script	Initial Supabase integration

5. Live Verification & Health Status

Supabase Cloud Connection: Verified active at https://nzccgolsbffelzdxtkom.supabase.co.
Unit Test Suite: Executed npm test with **65/65 tests passing (0 failures, 100% pass rate)**.
Data Integrity: All foreign keys, cascading deletes, composite primary keys, and RLS policies are enforced.

Report generated autonomously by Antigravity on behalf of Shaurya. ShopSense Project 2026.